Enhancing Next Activity Prediction in Process Mining with Retrieval-Augmented Generation

Appendix: complete per-log evaluation results

A. Casciani · M. L. Bernardi · M. Cimitile · A. Marrella
Information Systems, Elsevier (2026) · https://doi.org/10.1016/j.is.2025.102642
Companion document to the replication package · https://doi.org/10.5281/zenodo.21293307

Setup

This appendix reports one table per event log, with all ten evaluated LLMs and all metrics. Every table reports accuracy, macro-averaged precision, recall and F1 score, the total execution time in hours and the average per-inference time in seconds, for the ten benchmarked models: `Llama-3.1-8B-Instruct`, `Llama-3.2-1B-Instruct`, `Llama-3.2-3B-Instruct`, `Mistral-7B-Instruct-v0.2`, `Mistral-7B-Instruct-v0.3`, `Qwen2.5-7B-Instruct`, `Phi-4`, `DeepSeek-R1-Distill-Qwen-7B`, `DeepSeek-R1-Distill-Llama-8B` and `GPT-4o-mini`.

Results were obtained on a workstation running Ubuntu 22.04.3 LTS with an NVIDIA A100 GPU, using the single-bucket strategy and the `all-MiniLM-L12-v2` embedding model. Tables 1–7 cover the seven real-world logs and Tables 8–9 the two synthetic ones.

Reading the tables. Best value per column in **bold**, second best underlined, as in the published article these results come from.

Real-world event logs

Table 1. Next activity prediction on the *BPIC 2012* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
`Llama-3.1-8B-Instruct`	0.53	0.43	0.38	0.39	4.15	49.8
`Llama-3.2-1B-Instruct`	0.17	0.33	0.19	0.21	**0.85**	**10.3**
`Llama-3.2-3B-Instruct`	0.27	0.24	0.20	0.20	3.44	41.3
`Mistral-7B-Instruct-v0.2`	0.42	0.42	0.37	0.38	2.76	33.2
`Mistral-7B-Instruct-v0.3`	0.46	0.51	0.48	0.47	1.92	19.4
`Qwen2.5-7B-Instruct`	0.57	0.52	0.51	0.51	3.43	41.1
`Phi-4`	0.67	0.63	0.67	0.63	3.39	40.7
`DeepSeek-R1-Distill-Qwen-7B`	0.47	0.45	0.44	0.44	4.21	50.5
`DeepSeek-R1-Distill-Llama-8B`	0.55	0.55	0.53	0.54	7.78	93.3
`GPT-4o-mini`	**0.76**	**0.72**	**0.68**	**0.67**	1.68	20.1

Table 2. Next activity prediction on the *BPIC 2013 CP* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
Llama-3.1-8B-Instruct	0.72	0.70	0.71	0.70	3.54	42.4
Llama-3.2-1B-Instruct	0.17	0.17	0.16	0.16	1.21	14.5
Llama-3.2-3B-Instruct	0.28	0.26	0.27	0.26	1.96	23.5
Mistral-7B-Instruct-v0.2	0.42	0.37	0.37	0.36	2.98	35.7
Mistral-7B-Instruct-v0.3	0.66	0.55	0.57	0.55	**0.52**	**6.25**
Qwen2.5-7B-Instruct	0.83	0.78	0.79	0.78	3.52	42.1
Phi-4	0.89	0.85	**0.87**	0.86	5.96	71.5
DeepSeek-R1-Distill-Qwen-7B	0.62	0.58	0.57	0.57	7.43	89.16
DeepSeek-R1-Distill-Llama-8B	0.67	0.63	0.62	0.63	14.68	176.16
GPT-4o-mini	**0.90**	**0.87**	**0.87**	**0.87**	1.13	13.6

Table 3. Next activity prediction on the *BPIC 2013 I* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
Llama-3.1-8B-Instruct	0.37	0.33	0.32	0.32	5.28	63.3
Llama-3.2-1B-Instruct	0.03	0.03	0.02	0.03	2.65	31.8
Llama-3.2-3B-Instruct	0.05	0.06	0.05	0.05	3.05	36.5
Mistral-7B-Instruct-v0.2	0.38	0.31	0.30	0.30	4.49	53.8
Mistral-7B-Instruct-v0.3	0.12	0.12	0.11	0.11	4.38	52.5
Qwen2.5-7B-Instruct	0.45	0.32	0.32	0.31	2.71	32.5
Phi-4	0.53	0.57	0.53	0.54	5.15	61.7
DeepSeek-R1-Distill-Qwen-7B	0.39	0.38	0.37	0.37	5.69	68.2
DeepSeek-R1-Distill-Llama-8B	0.46	0.43	0.42	0.43	9.52	114.2
GPT-4o-mini	**0.70**	**0.64**	**0.65**	**0.64**	**1.79**	**21.5**

Table 4. Next activity prediction on the *BPIC 2020 ID* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
Llama-3.1-8B-Instruct	0.71	0.74	0.68	0.67	2.65	31.8
Llama-3.2-1B-Instruct	0.23	0.31	0.18	0.19	**0.71**	**8.5**
Llama-3.2-3B-Instruct	0.39	0.61	0.42	0.48	2.05	24.5
Mistral-7B-Instruct-v0.2	0.79	0.74	0.71	0.71	2.34	28.1
Mistral-7B-Instruct-v0.3	0.63	0.50	0.47	0.46	1.29	15.4
Qwen2.5-7B-Instruct	0.75	0.82	0.80	0.78	2.13	25.5
Phi-4	0.93	0.94	0.93	0.93	3.12	37.3
DeepSeek-R1-Distill-Qwen-7B	0.69	0.59	0.58	0.55	3.93	47.2
DeepSeek-R1-Distill-Llama-8B	0.70	0.70	0.69	0.67	7.34	88.1
GPT-4o-mini	**0.98**	**0.99**	**0.99**	**0.99**	0.92	11.0

Table 5. Next activity prediction on the *Hospital Billing* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
`Llama-3.1-8B-Instruct`	0.55	0.64	0.63	0.59	2.73	32.7
`Llama-3.2-1B-Instruct`	0.12	0.31	0.09	0.12	**0.86**	**10.3**
`Llama-3.2-3B-Instruct`	0.15	0.31	0.16	0.19	2.26	27.1
`Mistral-7B-Instruct-v0.2`	0.51	0.55	0.48	0.50	3.16	37.9
`Mistral-7B-Instruct-v0.3`	0.60	0.63	0.61	0.61	2.28	27.4
`Qwen2.5-7B-Instruct`	0.69	**0.72**	0.74	**0.72**	2.31	27.6
`Phi-4`	0.71	0.70	**0.78**	0.71	3.85	46.1
`DeepSeek-R1-Distill-Qwen-7B`	0.63	0.68	0.68	0.67	4.58	54.9
`DeepSeek-R1-Distill-Llama-8B`	0.56	0.55	0.58	0.53	9.58	114.9
`GPT-4o-mini`	**0.73**	0.71	0.74	0.71	1.19	14.2

Table 6. Next activity prediction on the *Sepsis Cases* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
`Llama-3.1-8B-Instruct`	0.53	0.60	0.60	0.58	2.98	35.7
`Llama-3.2-1B-Instruct`	0.18	0.39	0.23	0.24	1.05	12.5
`Llama-3.2-3B-Instruct`	0.26	0.41	0.27	0.31	2.41	28.8
`Mistral-7B-Instruct-v0.2`	0.50	0.66	0.59	0.61	2.74	32.8
`Mistral-7B-Instruct-v0.3`	0.55	0.67	0.68	0.64	**0.71**	**8.54**
`Qwen2.5-7B-Instruct`	0.66	0.80	0.79	0.79	2.87	34.4
`Phi-4`	**0.70**	0.77	0.76	0.77	3.47	41.6
`DeepSeek-R1-Distill-Qwen-7B`	0.55	0.56	0.60	0.54	6.86	82.3
`DeepSeek-R1-Distill-Llama-8B`	0.54	0.54	0.61	0.55	13.14	157.6
`GPT-4o-mini`	0.69	**0.85**	**0.83**	**0.84**	0.90	10.8

Table 7. Next activity prediction on the *Udon-ya* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
`Llama-3.1-8B-Instruct`	0.56	0.57	0.51	0.53	3.00	36.0
`Llama-3.2-1B-Instruct`	0.14	0.27	0.11	0.11	**0.71**	**8.4**
`Llama-3.2-3B-Instruct`	0.38	0.48	0.30	0.36	2.36	28.3
`Mistral-7B-Instruct-v0.2`	0.54	0.54	0.45	0.47	2.39	28.6
`Mistral-7B-Instruct-v0.3`	0.63	0.54	0.50	0.50	1.16	13.92
`Qwen2.5-7B-Instruct`	0.72	0.75	0.68	0.70	2.46	29.5
`Phi-4`	0.65	0.69	0.61	0.63	3.71	44.5
`DeepSeek-R1-Distill-Qwen-7B`	0.61	0.49	0.48	0.47	3.73	44.7
`DeepSeek-R1-Distill-Llama-8B`	0.58	0.55	0.50	0.51	6.89	82.6
`GPT-4o-mini`	**0.84**	**0.80**	**0.79**	**0.78**	1.37	16.4

Synthetic event logs

Table 8. Next activity prediction on the *SL2-2v-1r* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
`Llama-3.1-8B-Instruct`	0.39	0.17	0.16	0.16	0.33	25.8
`Llama-3.2-1B-Instruct`	0.09	0.14	0.02	0.03	0.17	13.3
`Llama-3.2-3B-Instruct`	0.13	0.14	0.06	0.08	0.24	18.8
`Mistral-7B-Instruct-v0.2`	0.18	0.10	0.09	0.08	0.50	39.1
`Mistral-7B-Instruct-v0.3`	0.20	0.16	0.11	0.11	0.33	28.8
`Qwen2.5-7B-Instruct`	**0.81**	**0.25**	**0.22**	**0.22**	0.39	30.5
`Phi-4`	0.72	0.14	0.14	0.13	0.67	52.43
`DeepSeek-R1-Distill-Qwen-7B`	0.69	0.14	0.13	0.13	0.50	39.1
`DeepSeek-R1-Distill-Llama-8B`	0.58	0.13	0.10	0.11	1.34	104.8
`GPT-4o-mini`	0.70	0.14	0.14	0.14	**0.13**	**10.4**

Table 9. Next activity prediction on the *SL3-MV* log.

LLM	Acc.	Prec.	Rec.	F1	Time	
					Total (h)	*Per-Inf. (s)*
`Llama-3.1-8B-Instruct`	0.25	0.10	0.09	0.09	2.40	28.8
`Llama-3.2-1B-Instruct`	0.07	0.07	0.01	0.02	**0.82**	**9.8**
`Llama-3.2-3B-Instruct`	0.42	0.06	0.06	0.06	1.89	22.6
`Mistral-7B-Instruct-v0.2`	0.23	0.11	0.07	0.08	3.44	41.2
`Mistral-7B-Instruct-v0.3`	0.22	0.11	0.08	0.09	2.82	33.8
`Qwen2.5-7B-Instruct`	0.46	**0.14**	**0.14**	**0.14**	2.44	29.2
`Phi-4`	0.42	0.12	0.10	0.11	3.83	46.0
`DeepSeek-R1-Distill-Qwen-7B`	0.42	0.12	0.10	0.11	2.94	35.3
`DeepSeek-R1-Distill-Llama-8B`	0.40	0.08	0.06	0.06	10.59	127.1
`GPT-4o-mini`	**0.47**	0.13	0.13	0.13	1.04	12.4